PROJECT REPORT / 01 NOVEMBER 2026

From run logs to crêpe insights

Maus Crêpe Insights | BaselHack 2026

Abstract. We built an offline-reproducible pipeline to turn Maus Robotics run records and camera checks into analysis tables, vision measurements, model artifacts and review tools. The public repository contains working components, but not a validated end-to-end robot system. Sparse check coverage and weak labels limit what the reported model scores establish.

828	177	138	62
run records	table columns	vision rows	human mask labels
exported as JSON and MCAP	one row per run	of 828 exported runs	valid comparisons

1 Problem and approach

Maus robots make crêpe-quality decisions from a camera above a hotplate. Steam, glare, browning, a rotating plate and tools crossing the view make single-frame inspection difficult. The challenge asks for spread quality, position and orientation, and fold state, with a path toward real-time use.

Dashboard → JSON / MCAP → Metrics + models → Review tools

The exporter writes one ROS 2 MCAP per run and a 177-column Parquet table. The vision pipeline reads the robot's annotated check images as measurements such as diameter, centre offset, holes and fold overlap. Shared frame preprocessing feeds compact ONNX classifiers. Separate labeling and mask prefill tools support future human review; an overlay renderer and run viewer make outputs inspectable. Private run data, videos and annotations remain out of Git.

2 What the measurements say

Evidence	Result	Interpretation
Coverage	138 of 828 runs have extracted vision measurements; 160 have any recorded check.	Coverage is selective, not a representative sample of all runs. Most failed runs lack a recorded check.
Segmentation student	Pixel IoU 0.966 against SAM 3 teacher masks on held-out runs.	Agreement with pseudo-labels, not a human-verified quality score.
Human comparison	Median mask IoU 0.944 on 62 valid masks; human–human median 0.933 on 10 matched frames.	Five consecutive runs from one robot; four were used for student training. This is a pilot, not an independent test.
Fold-state rule	99.5% agreement at 991 passed-check moments, versus 88.5% for the classical pipeline on the same frames.	Thresholds were read from these frames, including the test subset; results are optimistic.
Runtime	5.71 ms median preprocessing plus inference on a four-thread Intel CPU.	Raspberry Pi 5 estimate is 64 ms, not a measurement on the board.

Source: Guide: `crepe-student.md`; model card: `model.json`; coverage: `status.md`, snapshot d1c3701.

Readout. The project has a reproducible data and model toolchain and a promising offline segmentation prototype. Its human-label comparison is a small pilot, not an independent test; the on-robot decision path is not yet integrated.

A useful prototype, not yet a robot-ready system

3 Model results, with the right baseline

The strongest new result is a compact LR-ASPP MobileNetV3-Large segmentation student distilled from SAM 3 masks. It produces a crêpe mask; fold state is then estimated from mask area relative to that run's flat-crêpe reference. The exported ONNX model matches its PyTorch mask on the parity sample and is small enough for CPU inference. On the held-out runs, mask IoU against the teacher is 0.966. Check-moment agreement is 99.5% on 991 passed checks, but that figure is not an independent estimate: thresholds were selected using the same check moments, including the evaluation subset.

A first comparison against reviewed human labels reports median mask IoU 0.944 on 62 valid masks; two human label sets reach median IoU 0.933 on 10 matched frames. The label export covers five consecutive runs of one robot, and four runs also appear in student training. These numbers are a useful review pilot, not an independent estimate of performance on new robots or runs.

Earlier per-check classifiers predict the robot's own pass/fail verdicts, not verified food quality. Their holdout balanced accuracy ranges from 0.67 to 0.92 on only 31–61 frames per check; three of four cards say they do not generalize. These results are weak evidence for deployment and must not be read as human-validated performance.

4 Known gaps

- **Coverage:** vision measurements cover 138/828 runs, all from a short recent window in the documented export. Missing measurements are unknown, not evidence of a good or bad run.
- **Human evaluation:** the first comparison covers five consecutive runs of one robot; four runs were used to train the student. Broader, independent evaluation across runs and robots is still needed.
- **Scope:** the student has no hole class, does not estimate position or angle, and misses small fresh batter, hands and tools. It was trained on check-moment frames.
- **Generalization:** thresholds include evaluation frames; no leave-one-robot-out student evaluation has been run, and one robot has only six training runs.
- **Integration:** the repository has a mask model, review tools and overlay code, but no verified live path that returns all challenge decisions from one camera frame. Raspberry Pi performance is estimated, not measured.

5 What is reproducible

The code, model card, ONNX student, synthetic offline tests and documentation are public. The private dashboard export, frame tables, teacher masks and labeling records are not in the repository, so a fresh clone can reproduce the code checks but not the data-dependent counts or training run.

Offline check
`cd code && pixi install && pixi run test && pixi run lint`
This checks the public code; it does not rebuild private data artifacts.

6 Evidence trail

Claim	Repository source
Data counts and known gaps	`documentation/status.md` (snapshot revision d1c3701; counts from private export)
Student method and evaluation	`documentation/vision/crepe-student.md`
Model shape, splits and metrics	`code/models/crepe-student/model.json`
Model use and rebuild requirements	`code/models/crepe-student/README.md`
Project source	project repository

Evidence snapshot: public repository main at `f5d1504` (1 Nov 2026). Data coverage counts are explicitly tied to the older status snapshot because that page has not been refreshed for this revision. The model-specific results above come from the student guide and model card at the evidence snapshot. This report is a concise account, not a claim of production readiness.